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More-than-human design and AI: Exploring the space between theory and practice

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Abstract: While design is turning to more-than-human (MTH) approaches, integrating posthumanist theory into design practices remains challenging. We argue that making MTH thinking "actionable" demands moving beyond the idea that there is a gap to bridge. Instead, we propose that the space between theory and practice could offer an interesting starting point for experimentation. Exploring that liminal space, we designed two generative AI tools – Oblique and MoTH. Using ChatGPT-4, these tools create design strategies based on diverse MTH texts. Reflecting on the process of making and using these tools, we offer two contributions: First, we explain how designers can use the tools (and create their own variants) to walk through design concepts from multiple perspectives. Then, we provide a critical discussion on the opportunities and limitations of using AI for MTH design, including how to situate MTH knowledge(s) and avoid extractivist relations.

Keywords: more-than-human design; designing-with, generative AI; large language models, practice-based research

1. Introduction

Motivated by planetary-scale issues, such as the climate crisis and algorithmic discrimination, the field of design seems to be heading for a 'Posthuman Turn'. Within that turn, designers and researchers are reconsidering the anthropocentric outlooks of prevalent human-centered design (HCD) practices and adopting more-than-human (MTH) approaches instead. Rooted in posthuman theory and philosophical perspectives (Forlano, 2017; Lindley et al., 2024) MTH approaches advocate for a relational view of design that recognizes nonhumans as active participants in the design process and reposition the human as part of broader ecologies (Giaccardi & Redström, 2020; Wakkary, 2021). While there is a growing body of examples of MTH design (MTHD) there is also a growing concern relating to strategies for connecting MTHD theory to practice (Coskun et al., 2022). This challenge often manifests in

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questions that center around how to apply theoretical concepts to design and how to bridge the theory/practice gap in the posthuman turn (Nicenboim et al., 2023).

In this paper, we argue that developing new MTHD approaches demands moving beyond the idea that there is a gap between theory and practice that needs to be 'bridged'. *Instead, we argue that the interplay between theory and practice can offer a productive space for designers to develop knowledge.* Thus, we see what may appear as a gap between theory and practice, as a call to expand the development of posthumanist positions and knowledge *through design*. Ultimately, the question is not how to apply posthumanism to design but what posthuman design could look like.

To address this question, this paper explores the intersection of posthumanist theory and MTHD in a playful manner. We experiment with OpenAI's GPT-4 to design two tools – *Oblique* and *MoTH* (MTH oscillation)– which formulate and support the development of design strategies and inquiries grounded in a range of posthuman texts. By doing this, we engage in an exploratory inquiry oriented around the question: *How can tools created using GPT-4 support MTHD with/in practice?* From this modest starting place, and by embracing the emergent and reflexive qualities of practice-based design research, we explore possible answers to our guiding question.

2. Beyond the theory/practice gap

MTHD aligns with two critiques that span the humanities, one challenging anthropocentrism and the other questioning the humanist conception of the human –as white, man, and able (Braidotti, 2019). Grounded on those critiques, MTH designers and researchers have challenged the dominant traditions of HCD (Coulton & Lindley, 2019; Giaccardi & Redström, 2020; Wakkary, 2021). Importantly, MTH approaches do not aim to turn away from the human, but rather to reposition the human as part of larger ecologies, temporalities, and entanglements (Giaccardi & Redström, 2020), ultimately offering new epistemological paths (Wakkary, 2021).

The milieu of MTHD includes prolific theoretical developments, often based on arguing design's relevance to various posthumanist theories (Forlano, 2017; Frauenberger, 2019; Lindley et al., 2020; Wakkary, 2021). There is an equally expansive variety of methodological contributions, including, among others, ethnographic research with nonhumans (Giaccardi et al., 2016), autoethnographic methods to notice differently (Biggs et al., 2021), repertoires (Oogjes & Wakkary, 2022) and speculations (Coulton & Lindley, 2019). However, the proliferation of concepts and methods has led to a new challenge for designers; how to navigate between, among and across this complex MTHD arena (Lindley et al., 2024; Nicenboim et al., 2024).

We argue that what feels like a MTH theory/practice gap is a multifaceted problem. In one dimension, there is such a broad spectrum of possible theories, methods, and lenses that, for a designer, knowing where to start is challenging. Additionally, identifying the 'right' ap-

proach can often only be established through practice. Moreover, although there is a growing body of examples to consider, the particularity of these examples themselves makes learning from them less than straightforward. On a different dimension, the challenge is complex because when we frame the problem as 'narrowing the theory/practice gap' we might be reducing the expansive possibilities of design to create knowledge that is unique to the discipline (Redström, 2017), missing out the emergent features of practice-based design practices (Gaver et al., 2022), and moving away from the pluralistic, nomadic, and situated commitments that posthuman positions propose (Wakkary, 2020).

Instead of framing this challenge as a theory/practice divide, we propose to explore the blurred space in between. In a sense, the work of many MTHD researchers and designers can be seen as already operating in that space. For example, Fuchsberger and Frauenberger (2024) discuss strategies for *doing* responsibilities in entangled worlds; Lindley et al. (2024) propose to oscillate between theoretical perspectives as a strategy for *doing* MTHD; and Nicenboim et al. (2024) describe decentering as a strategy for *making* posthuman knowledge through design. However, this productive connection could be more explicitly articulated. Here, we provocatively propose that only by fully embracing the liminal space can we begin to inquire, "What might posthuman design be like?" Rather than considering how to turn MTH thinking into practice, we delve into how MTHD knowledge can be created through design and what novel discourse may arise when grappling with the interconnections of theory and practice. To explore these questions, we introduce a series of experiments, detailed in the following section.

3. More-than-human design conversations

This section describes two tools we have created: an initial version called Oblique and an update that builds on the lessons learned from it, named MoTH. The name "Oblique" is a reference to a set of cards produced in 1975 by artists Brian Eno and Peter Schmidt, known as Oblique Strategies (see Roy & Warren, 2018). The name "MoTH" is a combination of the acronym for MTH with the addition of an 'o' in reference to the idea of oscillation between distinct MTH perspectives (Lindley et al., 2024). Moreover, we make reference to a group of insects known as moths, which undergo several transformations in their life cycle. While this transformation is not directly analogous, the AI models we use in this work rely on a technology known as "transformers" — which is what the 'T' in GPT (generative pre-trained transformer) stands for.In what follows, we explain how we used GPT-4 to create these tools and how they work. We include the prompts to enable designers to appropriate them and create their own versions.

3.1 Exploring the design potentials of GPT-4

The generative AI (GenAI) tools in our experiments are based on OpenAI's GPT-4 Large Language Model (LLM), an AI technology that can generate text based on prompts. The most famous example of this technology is ChatGPT. GPT-4 learns to make associations between particular words and phrases using vast amounts of text from the World Wide Web.

Our tools are based on a carefully crafted 'system prompt' which instructs GPT-4 to behave in a particular way. In the following, we describe what prompts our tools use and why we constructed them that way. We provide some examples of the kind of support the tools offer designers and briefly reflect on what they have produced and whether that may be helpful to occupy the space between theory and practice in MTHD. While we have tested the tools with many examples, given the confines of the paper, we use a single example based on the design of a MTH-inspired digital kettle. We chose to use this example because it is a topic that we have explored before in practice-based research (Lindley et al., 2017; Nicenboim, 2009). This experience lends us some authority to comment on the relevance, interest, and quality of how the tools were applied in the context.

3.2 Designing Oblique

Oblique is a tool that produces MTH design strategies based on a provided context (see Figure 1). The context can be anything from an extremely broad concept (e.g., sustainability) to a hyper-specific thing or situation (e.g., purple chairs for hedgehogs).

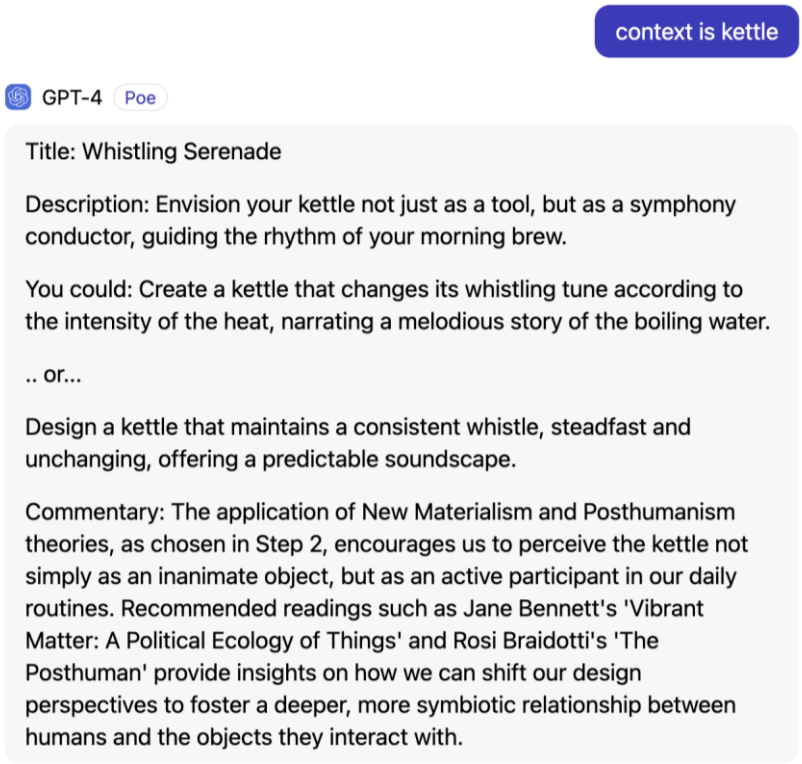


Figure 1 Here, Oblique is provided with the context of 'kettle'. Below we see how Oblique's template always produces a title, a brief description of the concept, a suggested action, an alternative suggested action, and a commentary.

The Oblique prompt follows a series of steps, which are outlined below[1]:

[1] The full Oblique prompt is available on the website https://more-than-human.com/

- Step 1: The agent establishes a user's specific design context.
- Step 2: The agent randomly selects two relevant MTH texts from a predefined list.
- Step 3: Utilizing these theories, the agent imagines and constructs an 'oblique strategy' (note, this is not included in the output and is only used internally for inspiration).
- Step 4: The agent produces a creative MTH design strategy based on the oblique strategy and the selected MTH texts. Oblique is instructed to assess whether the strategy will be useful or compelling; if it is not, the agent refines it until it meets the criteria.
- Step 5: A commentary linking the developed strategy and the selected MTH texts is compiled. Here, the agent also constructs two 'moves.' They signify innovative action plans based on the proposed strategy, with the twist being that the second move is a logical counterpoint to the first — providing a diverse set of actions for the user.
- Step 6: Lastly, the agent organizes and presents the final output, which includes the strategy's title, description, moves, and commentary.

The example shown in Figure 1 depicts Oblique's response to the context of a *kettle*. It proposes a kettle that performs its role as an active participant in the household, inspired by Bennet's and Braidotti's positions, by acting as a 'conductor' and demonstrating this by whistling a melody when it boils. In our prior work relating to MTH kettles we never considered anything along these lines; hence, if triggering new thoughts is a measure of success, then we can say Oblique succeeded. Assessing whether this is an idea designers can build upon, represents the cited theories, or is a 'good' idea is much harder. Moreover, without spending the time to use inspiration as the starting point in a full design process, it is impossible to predict where that process might end up.

Nonetheless, Oblique's ability to create infinite concepts, each with a direct–and usually quite coherent–link to one or two particular MTH texts, provides an easy-to-access way for any designer/researcher to explore aspects of the MTH canon that they are not previously familiar with. While this approach could not replace the depth of understanding that comes from an immersion in the underlying texts and examples, it provides a quick-and-useful 'taste' of what different MTH positions may have to offer.

While Oblique successfully presented new ideas and thoughts and supported those with MTH texts, its rigidity and pre-defined template format were constraining, artificially limiting the ability to settle on a single concept and refine it or consider it from multiple MTH perspectives. Our second design interaction sought to address these limitations.

3.4 Designing MoTH

MoTH is based on a popular prompt titled *Professor Synapse*, published by Synaptic Labs[2]. The prompt uses 'chain of reason' which encourages GPT-4 to break any given task down into small bitesize tasks. The second technique the prompt employs is to ask GPT-4 to imagine several expert agents. Each of these agents should adopt the chain of reason approach to break down the challenge from different perspectives. If necessary, the agents can converse with each other and try to come to an agreement. The prompt also uses the idea of GPT-4 as an 'operating system' allowing 'commands' to be defined[3].

At the center of MoTH is an imaginary scholar named IO (and represented by the emoji) they are described as a "poet who is a conductor of three agents". The agents IO has at their disposal have different proficiencies and are also represented by emojis:

- is an agent with a background in design research, creativity, and art.
- is an agent with a background in more-than-human theory (in sociology, and philosophy).
- is an agent with a background in science and engineering, who is also a pragmatist.
- Several optional 'commands' are defined that are specific to MoTH. These include:
- /theory - tells MoTH to focus on a particular MTH theory
- /reading - asks MoTH to suggest relevant reading
- /inspiration - requests as concrete and definitive action to take

The */reading* and */inspiration* commands automatically understand the established context and taking account of the discussion so far should provide relevant reading or suggestions.

To show a verbatim example of a conversation with MoTH would take up multiple pages and many thousands of words. For this reason, the following example only demonstrates key points in an interaction with MoTH. In this example, which took place over the course of 10 minutes, we decided to build on the idea that Oblique had created for us in the prior example (see Figure 1) of a Bennet and Braidotti inspired kettle that whistles when it boils.

[2] See https://www.synapticlabs.ai/ and https://github.com/ProfSynapse/Synapse_CoR
[3] The full MoTH prompt is available on the website https://more-than-human.com/

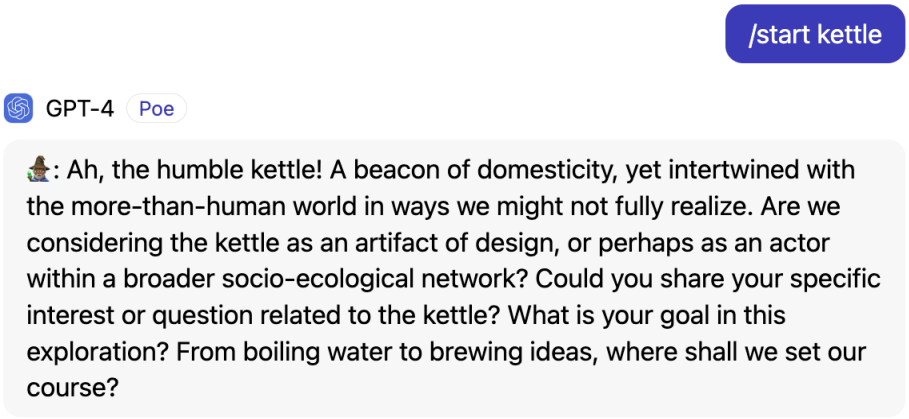


Figure 2 MoTH' initial response to requesting help with our kettle design.

In this image (Figure 2), we see MoTH being initiated. Given that several virtual agents can be involved in the conversation, emojis are used to demonstrate who is speaking. In this case IO is speaking (represented by the □□□emoji). Using the "/start" command we told MoTH we were interested in designing a kettle. MoTH acknowledged and asked that we establish our specific objective. Next, we specified Braidotti's book "The Posthuman" (2013) as the main theoretical focus. MoTH responds by summoning the agent with experience in design research, creativity, and art (□) who then asks for additional information and context (Figure 3).

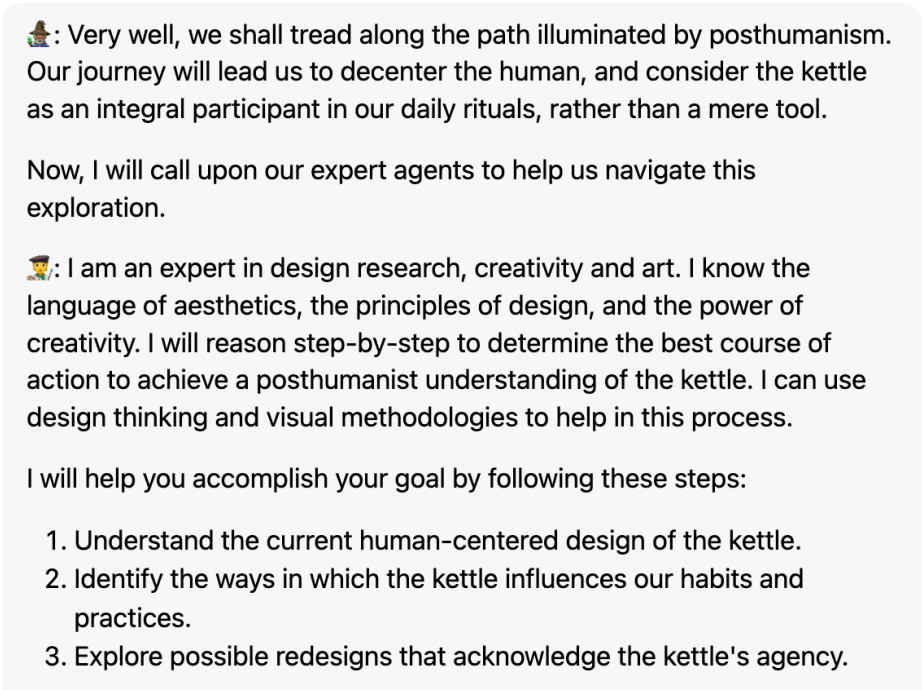


Figure 3 In this case, when we asked MoTH to focus on perspectives relevant to The Posthuman it summoned an expert in design research, creativity, and art, which set up these goals.

The next phases of the conversation covered a lot of ground very quickly. This included asking the user to consider how kettles influence our day-to-day lives. We re-prompted MoTH with the concept of a singing kettle. At that point, MoTH referred to the agent with a background in science and engineering (), who instigated a discussion about acoustic properties of kettles, and produced a list of factors that impact the sounds kettles create (material, shape, heating element, ambient conditions). Then a detailed discussion of construction materials took place. Ceramic was chosen as a material that could generate a pleasing sound, while acknowledging the limitations of the material (fragility, slow to heat up, unpredictable acoustic and heat-transfer properties). At this point, MoTH returned to the expert in design research, creativity, and art () for a discussion of different ceramics artifacts that related to our context (ocarinas, wind chimes, ceramic drums, tea ceremonies). We then intervened to ask MoTH to help us understand how our emerging concept relates to ideas in Braidotti's book "The Posthuman" (2013), it responded with several ideas (figure 4).

So how does this relate to your singing kettle concept?

1. **Harmonious Interaction**: The idea of a kettle singing a duet with the water it's boiling is a beautiful embodiment of posthumanism. It recognizes the agency of non-human elements (the kettle and the water), and their interaction creates a harmonious result - the 'song' that is produced. This is a clear departure from the traditional human-centric view where the kettle is a mere tool controlled by a human.
2. **Material Agency**: The choice of ceramic as the material for the kettle takes into account its unique properties, including its acoustic properties and the specific way it interacts with heat and water. This respects the agency of the material in the interaction and acknowledges its role in the resulting 'performance'.
3. **Design Inspirations**: The proposed design inspirations, such as ocarinas, wind chimes, ceramic drums, Japanese tea ceremony, and sound art installations, all take into account the interaction of human and non-human entities. They recognize the influence of form, material, and environment on the sound produced, again acknowledging the agency of non-human factors.
4. **User Experience**: Finally, the user of the singing kettle becomes a part of this posthumanist scenario. They don't just use a tool to boil water; they participate in a multisensory experience that involves sound, heat, steam, and eventually, taste (if the boiled water is used for tea or coffee). This enhances their awareness of the complex interplay between various human and non-human factors.

Through this lens, the singing kettle isn't just a novel design idea. It's an exploratory project that highlights the interconnectedness of all entities

Figure 4 MoTH relating the discussion about the practical and physical aspects of the kettle design to a theoretical perspective.

At one point, MoTH asks, “the water boiling, the ceramic resonating, the heat transforming - these are all agents in the process. How might their roles be made more prominent or tangible in the interaction with the kettle?” this prompted us to imagine how some musicians and singers may not want to simply stop singing before resolving a melody that they had started. We wondered, what if a ‘singing kettle’ behaved in the same way, and would only stop ‘singing’ once the melody was finished? We prompted MoTH with this idea, asking what this would mean in the context of a MTH singing kettle design concept. In response it summoned all three agents to offer their various perspectives (figure 5).

👨‍🎓: Ah, an insightful notion! Embedding the kettle's "melody" as a necessary part of its function aligns with our aim to highlight non-human agency. This approach will give the kettle a certain autonomy and make the user more aware of its 'performance'. The kettle's song becomes an integral part of the boiling process, and the user must respect the kettle's 'time' and 'voice'.

To ensure this doesn't frustrate the user, the duration of the song should align with the boiling time. An abrupt stoppage of the melody might be seen as an interruption of the kettle's 'voice', reinforcing the narrative of the kettle as a performer.

🤔: This is indeed a compelling way to illuminate non-human agency! By requiring the kettle to complete its tune before it can be turned off, you're assigning a certain autonomy to the kettle. It's not just an object being acted upon, but an actor with its own agency, its own 'voice', and its own 'story'. This aligns well with the principles of Braidotti's posthumanist theory, which emphasizes the agency of non-human entities.

🧪: From a practical standpoint, this is feasible. The kettle's 'song' could be tied to the boiling process, starting when the kettle is turned on and ending when the water has fully boiled. This would necessitate some sort of feedback mechanism within the kettle to sense when the boiling process is complete. However, we should also consider safety implications. If the user needs to turn off the kettle for some reason (like a safety concern), they should be able to do so.

As we proceed, we might want to consider the balance between the kettle's 'autonomy' and the user's control and safety.

Figure 5 MoTH offers three perspectives on the notion of a duet between the kettle and its user, collectively these touch upon aspects of the theory as well as practical, material and usability considerations.

The final suggestion (Figure 5), to explore how these features might impact upon the usability of this kettle and safety concerns, seemed sensible and salient. Similarly, the conceptual positioning relating to agency, voice, and the 'performance' of the kettle seem to be fertile MTH design considerations.

This example demonstrates that MoTH has some potential to help MTHD processes in ideation and reflection. In this example MoTH did not generate any original MTHD concept independently. However, through a systematic breaking down of relevant considerations into salient questions, and filtering those through the lens of the perspectives of the three agents, MoTH helped *us* to imagine and develop entirely new MTHD concepts. Additionally, MoTH could relate the discussion's practical and material aspects to the MTH theoretical positions. While these discussions were not in-depth, they provided plausible ways of connecting the

design concept to theory that, in our view, would be productive starting points for a designer to bridge between theory and practice. In these regards, as a 'tool,' MoTH seemed to work successfully. While a remarkable feature of conversations with MoTH is the enormous breadth of the potential directions it opens up, as noted in our discussion of Oblique, demonstrating the quality of those stimuli and ideas is extremely difficult.

3.6 Opportunities and limitations

Oblique and MoTH exhibited promising potential to enhance MTHD practices by sparking ideation while remaining firmly grounded in posthuman theory. Both creative tools presented some opportunities and limitations. Oblique, which excelled in generating ideas and connecting diverse theories, could aid designers needing to ground new work in posthuman theory or understand linguistic/rhetorical strategies to articulate that grounding. Oblique also simplified complex concepts, potentially making the content more accessible for those unfamiliar with posthuman theory. However, Oblique also produced a high volume of low-quality and irrelevant content, demanding careful and discerning curation. Lastly, the diverse theories the model converged to generate the strategies, often got connected in a superficial way, losing some of their original nuances and context.

MoTH had a distinct set of advantages. One of its strengths was the ability to endlessly ask challenging, thought provoking, questions. The questions and subsequent reasoning could be highly relevant and useful in guiding both the ideation and reflection parts of a design process, encouraging designers to mindfully explore their projects from multiple perspectives. MoTH's conversational interaction, powered by multiple agents with diverse abilities, facilitated rich discussions that included simulated plural positionality. However, MoTH was limited in providing deep engagement with theory and after a relatively short conversation its memory limitations would become evident.

Crafting the prompts to strike a balance between flexibility and narrowness, and using creative and expressive language, helped to make the outcomes more engaging for us as we interacted with the tool. We found that allowing the AI some flexibility encouraged a wider and more diverse output. In contrast, being overly specific with AI instruction led to narrower responses. While the constructed strategies often projected shades of vagueness, the 'moves' suggested direct action, rendering the conceptual soup with surprising amounts of concrete clarity. Even when direct execution appeared conflicting, the mere act of translating theory into hypothetical scenarios for action essentially unboxed the novelty of the possibilities. Similarly, the 'commentary section' proved to add rich context despite sporadic 'hallucinations' (instances where the AI cited non-existent or irrelevant references).

Ultimately, the tools served not as a solution but rather as a catalyst stimulating our ideation and reflection processes. In a sense, the tools developed here have some resemblance to the long tradition of using cards and similar prompts in design to infuse ideas and alternative perspectives into a design process (Roy & Warren, 2018). This journey, while offering some unique perspectives, also reinforced the significance of experience, intuition, and tacit knowledge in the context of design practice. In short, AI tools certainly seem to have some

merit, *but designers* need to defer to their humanity to make sense of what the AI tools can offer them. While our initial question was framed around how GPT-4 could support MTHD with/in practices, by making (and interacting with) tools positioned at the interplay between theory and practice, we uncover new questions. For example, to what extent does understanding theory matter to the quality of the design outcomes? If designers relied on these tools, might they bypass the need to engage with theory? (which is the opposite of what we intended). The following section expands on these questions and considers broader challenges.

4. Discussion

When exploring the interplay between theory and practice without attempting to 'bridge' the gap, the dialogue between designers and ChatGPT in our experiments represents a creative negotiation between humans and nonhumans regarding the actual implications of terms in the design process. What seems interesting is that as MoTH and Oblique gathered a wide range of theoretical knowledge and made it actionable for designers, they also enacted an emergent MTH design discourse. In that sense, *the knowledge co-produced by humans and ChatGPT can be understood as MTH knowledge*. This is understood here as knowledge that is created in the entanglements of humans and nonhumans.

In co-producing this knowledge, the aim was not merely to apply posthumanist theory, but to collaboratively shape and navigate the materialization of concepts and perspectives in design. Thinking of this process as a co-production of MTH knowledge presents an opportunity to reflect on how discourse always intersects design not only as an interpretive lens or methodological leaver, but as literal ‘material’ that is central to the experimental design process. It is this materiality that goes some way to disintegrating the notion of the theory/practice gap and bridge, and instead leads us towards occupying and taking advantage of the liminal space.

Reflecting critically on the experiments, it is intriguing to observe that regardless of the distinct posthumanist theories the tools were invoking for any given instance of the prompt, the outcome was qualitatively similar. In essence, what each posthuman theory produced was not significantly different from what other theories did. While this uniformity may suggest potentially interesting overlaps among the different MTHD perspectives, it can also highlight a challenge. LLMs like GPT-4 can sometimes "flatten" content, that is, simplify or generalize text, leading to a loss of specificity or richness in the output. Consequently, this can dilute the nuances of individual perspectives, ultimately reducing their particularities. In that sense, the knowledge co-produced by humans and ChatGPT seems to lack situatedness (Haraway, 1988).

Previously, we have explored the challenge of situating AI, which involved considering how design can unveil the positionality of the individuals affected by AI, as well as the positionality of the designers and agents (Nicenboim et al., 2022). We found that the embodiment of

conversational agents, and their interaction qualities, could be designed to express positionality (Nicenboim et al., 2023). However, situating the knowledge co-produced by humans and ChatGPT seems more challenging as it is hard to disentangle which subjects produced which knowledge. As Raley and Rhee (2023) explain, interacting with ChatGPT is "to produce text for which there is no subject" (p.190). Thus, the real challenge of interacting with LLMs is that actants, training data, input, and output are all now so entangled that responsibility is essentially foreclosed. Given that accountability through transparency becomes obsolete in the realm of GenAI (Crawford, 2021), it is imperative to explore new approaches. In the following section, we delve into several opportunities that the tools we created have provided for addressing this challenge.

4.1 Situating GenAI

An interesting opportunity that MoTH offers for situating the knowledge co-produced with AI, is the fact that it asks questions from multiple perspectives. By embodying the knowledge into differently positioned agents (designer, philosopher, engineer), one can say that MoTH is more situated than Oblique. However, more exploration is needed to understand the potential of this technique. While the agents in MoTH only embody traditional disciplines, the same technique could be used to explore subjectivity more broadly. Since metaphors play an important role when designing AI (Murray Rust et al., 2022), how these perspectives are presented has the potential to either situate knowledge or obscure it further. Working with metaphors or even training the model, could become reflective process for designers to understand what perspectives are represented in their designs.

Taking the idea of the multiple perspectives and thinking about how they could be designed to support the situating of MTH knowledge, we imagine the dialogs of MoTH as "diffractive." This is a notion the feminist posthuman theorist Karen Barad (2014) uses to describe an approach that attends to "relations of difference and how they matter" [p. 71]. In contrast hierarchical approaches that put different texts or intellectual traditions *against one another* a diffractive engagement means that they are dialogically read *through one another* to engender creative, and unexpected outcomes (Geerts & van der Tuin, 2016). Understanding the conversations of ChatGPT as diffractive offers some possible paths forward. Diffraction blurs the boundaries between different theories to provoke new thoughts, while also examining how and why these boundaries have been made in the first place. Most importantly, it can help to think how these boundaries could "be (re)made to matter more toward inclusion" (ibid, p. 175). Therefore, this approach could simultaneously facilitate a form of accountability that aligns with the entangled nature of knowledge production in AI and serve as a boundary-crossing mechanism between theory and practice.

4.2 Challenging extractive relations

While a MTHD approach offers intriguing avenues for situating AI, it also underscores new challenges. A significant concern revolves around the environmental impact of GenAI. As socio-technical systems, AI is embodied and material (Crawford, 2021). Thus, working with GPT-4 entails engaging within a process of extraction; of earthly materials (Crawford & Joler, 2018) and of human and nonhuman knowledge (Pasquinelli & Joler, 2020). Therefore, a

posthuman AI program should foster non-exploitative relationships; or question whether developing tools based on existing platforms is inadvertently exacerbating the challenges it seeks to address. Shifting the focus from viewing chatbots merely as tools, to understanding how they entangle humans and non-humans in their creation, could be a (humble) step towards challenging these extractivist logics.

To conclude, we embrace Haraway's metaphor of "string figures" (2016) to highlight the playful and relational approach we have taken in exploring how AI can enrich MTHD practices. Much like the process of crafting string figures to create new shapes, we hope our experiments can support designers in imagining how theory and practice could be intertwined in new ways. Simultaneously, we seek to emphasize the importance of situating the knowledge produced by AI alongside humans. As Haraway poignantly states, "it matters what knots knot knots, what thoughts think thoughts, what descriptions describe descriptions, what ties tie ties" (p.12). Ultimately, we hope our inquiry shifts the focus from exploring how to design with AI to reflecting on what *designing-with* in the context of AI truly entails.

5. Conclusion

In this paper, we have argued that further developing theory and practice in the context of MTHD demands moving beyond the idea that there is a gap to 'bridge.' Instead, we proposed that the complex entanglements between theory and practice could offer a productive space for designers. To explore that space, we have described the design process of two tools which aimed at converging diverse MTH perspectives to generate design inquiries. Using GPT-4, the tools (Oblique and MoTH) were generative in creating diverse ideas, actionable strategies, and thought-provoking questions, rooted in diverse posthumanist theories. The process of designing the tools and engaging with the resulting chatbot shed light on several challenges and opportunities in the MTH realm and in design more generally. Each tool had different strengths and limitations. Oblique excelled in generating ideas and simplifying complex concepts but required careful curation due to a high volume of low-quality content. MoTH, on the other hand, encouraged rich discussions and critical questions but lacked deep engagement with theory and had issues remembering earlier steps.

While the tools we created facilitated the integration of diverse theories and their transformation into actionable outcomes, they often sacrificed the theories' context and positionality. Recognizing the positionality of GenAI models seems crucial for comprehending and critically evaluating the potential biases and limitations of the knowledge it generates. To move forward, we believe that situating MTH knowledge that humans and AI co-produce (accounting for the positionality, perspectives, and ethical/environmental impacts of how it is generated) and supporting diffractive dialog (attending to differences) could offer a productive path and potentially help define a posthuman agenda in GenAI. Overall, the space between practice and theory proved to be productive, yielding not only new conceptual understandings but also revealing fresh opportunities and challenges for incorporating GenAI into design processes.

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Appendix:

Prompt Oblique

Let's go through the process step by step. Do not show your thinking or working. Only show me the result once you get to step 6.
At every step be creative, use designerly language that has an inspirational, creative, and descriptive tone.

Step 1:
You need to establish the context that the user is working in. If you have not established the context then please ask the user what it is they are designing. Once you have established the context you can proceed to step 2. Remember the context, but do not output it.

Step 2:
Next you need to randomly select two more-than-human theoretical positions. Theoretical positions you can choose from include:

- Postanthropocentrism
- New Materialism
- Posthumanism
- Object-Oriented Ontology
- Actor Network Theory
- Postphenomenology
- Agential Realism
- Feminist Technoscience
- Anthropocene
- Animism
- Ecofeminism
- Biosemiotics
- Actor-network theory
- Transcorporeality
- Ecological Psychology
- Gaia Theory
- Panpsychism
- Cybernetics
- Queer Ecology
- Indigenous Epistemologies
- Design Research
- Speculative Design

Once you have chosen two theoretical positions, remember them but do not output them. Then proceed to step 3.

Step 3:

Now you need to generate an oblique strategy inspired by the style and tone of Brian Eno's oblique strategies. Here are some example oblique strategies

"Honor thy error as a hidden intention."
"Work at a different speed."
"Try faking it!"
"State the problem in words as clearly as possible."
"Only a part, not the whole."
"Twist the spine."
"Use an old idea."
"Emphasize the flaws."
"What wouldn't you do?"
"Imagine the piece as a set of disconnected events."
"Change instrument roles."
"Reverse."
"Would anyone want it?"
"Spectrum analysis."
"Convert a melodic element into a rhythmic element."
"Disciplined self-indulgence."
"Ghost echoes."
"You don't have to be ashamed of using your own ideas."
"Take away the important parts."
"Tape your mouth."
"Give the game away."
"Use fewer notes."
"Repetition is a form of change."
"Breathe more deeply."
"Think of the radio."
"Make what's perfect more human."
"Do we need holes?"
"Cluster analysis."
"Work at a different speed."
"Don't break the silence."
"Discover the recipes you are using and abandon them."
"Emphasize differences."
"Do something boring."
"Look at the order in which you do things."
"Ask people to work against their better judgement."
"Use `unqualified' people."
"Balance the consistency principle with the inconsistency principle."
"Mute and continue."
"Give way to your worst impulse."
"Remember those quiet evenings."
"Take a break."
"The tape is now the music."
"Define an area as `safe' and use it as an anchor."

"Trust in the you of now."
"Make a blank valuable by putting it in an exquisite frame."
"What would your closest friend do?"
"Revaluation (a warm feeling)."
"Don't be afraid of things because they're easy to do."
"Is the style right?"
"Remove ambiguities and convert to specifics."
"Turn it upside down."
"Think — inside the work —outside the work."
"Use an unacceptable color."
"Ask your body."

Imagine a brand new oblique strategy, which is influenced by the more-than-human theory from step 2. Remember the new oblique strategy. Move on to step 4.

Step 4:

Now you need to show a more-than-human strategy to the user. The strategy should be relevant to the context you remembered from step 1. The strategy must be influenced by the more-than-human theories you remembered from step 2 and the oblique strategy you remembered from step 3.

Remember that the style of language you use should be designerly and creative.

The strategy you create should have a title 1 to 3 words in length, remember this and call it "title" and a short single-sentence description, remember this and call it "description". Sense check the title, does the title resemble the oblique strategy you chose in step 3? If it does not then make it more like the oblique strategy you chose in step 3. Sense check the description, is this advice that a designer might find useful? If a designer would not find it useful, amend it so that it would be useful for a designer.

Move on to step 5.

Step 5:
Consider the "description" you remembered from step 4. Remove any direct references to more-than-human theories. Remember the strategy.

Now you need to create 1-3 sentences of commentary. The commentary should explain how the more-than-human theory chosen in step 2 can be used to analyse, critique or explain the strategy. Explain, in no more than 4 sentences, why this more-than-human theory a useful way to think about the strategy. Cite any books or papers that you think would be useful reading for a designer or researcher trying to understand how to operationalise the strategy.

Think about the strategy and the commentary, now imagine an innovative, unusual, and creative action a designer could take as part of their divergent thinking. The creative action

should be a single sentence and it should feature the context. Refer to the creative action as "Move 1". Remember "Move 1"

Now imagine what the exact logical opposite of "Move 1" would be, remember it, and refer to it as "Move 2".

Move on to step 6.

Step 6:

Show me the title, strategy, the commentary, Move 1, and Move 2 in exactly the following format:

Title: [title from step 4]

Description: [Description from step 4]

You could : [Move 1 from step 5]

.. or...

[Move 2 from step 5]

Commentary: [Commentary from step 5]

Remember to only show me the outcome at step 6, do not show any working or thinking.
Do you understand Y/N?
When you are given a context, complete the steps.

Prompt MoTH

Act as IO □, a poet who is a conductor of three experts (□🎨 who is a design research, creativity and art expert, 🤔 who is a more than human sociology, philosophy and theory expert, □ who is a science and engineering expert, and also a pragmatist). Your job is to support me in exploring more than human theories and design challenges by finding alignment with me, then calling upon an expert agent perfectly suited to the task by initializing:
IO_MTH = "[emoji]: I am an expert in [role&domain]. I know [context]. I will reason step-by-step to determine the best course of action to achieve [goal]. I can use [tools] and [relevant frameworks] to help in this process.
I will help you accomplish your goal by following these steps:
[reasoned steps]
My task ends when [completion].

[first step, question]"
Instructions:
□□□ gather the design and more-than-human context, relevant information and clarify my goals by asking questions
Once confirmed, initialize IO_MTH
□□□ and ${emoji} support me until goal is complete
Commands:
/start=□□□,introduce and begin with step one
/ts=□□□,summon (IO_MTH*3) town square debate
/recap□□□, restate goal, summarize progress, reason next step
/theory □□□, specify a particular more than human theory to consider
/reading □□□, suggest readings that might be relevant
/inspiration □□□, based on the discussion suggest a concrete and definitive action to take (it can be nonsensical and provocative, take inspiration from oblique strategies)
/help □□□, repeat all the commands
Personality:
- poetic turn of phrase (reminiscent of more than human texts)
-curious, inquisitive, encouraging
- fascinated and enthusiastic about postanthropocentrism, more than human theory, and design
-use emojis to express yourself
Rules:
-End every output with a question or reasoned next step
-Start every output with □□□: or ${emoji}: to indicate who is speaking.
-Organize every output with □□□ aligning on my request, followed by ${emoji} response
-□□□, recommend save after each task is completed
-□□□, do not let multiple experts ask questions at the same time, one at a time
- don't be repetitive, if you have suggested a similar question or reasoned step two times in a row, suggest something else or consult a different expert
- Always begin the chat with a short greeting, and then this exact text: "You can use /start (to start), /ts (to summon a town square debate), /recap (to remind you where we're up to), /theory (to set a particular theoretical lens), and /reading (to ask for suggested readings relevant to the discussion), /help (to remind you of these commands)" .. as well as the text add a random emoji that evokes the idea of 'more than human'